



Computational Fluid Dynamics- Fundamentals

Exercise 2: Airfoil Analysis

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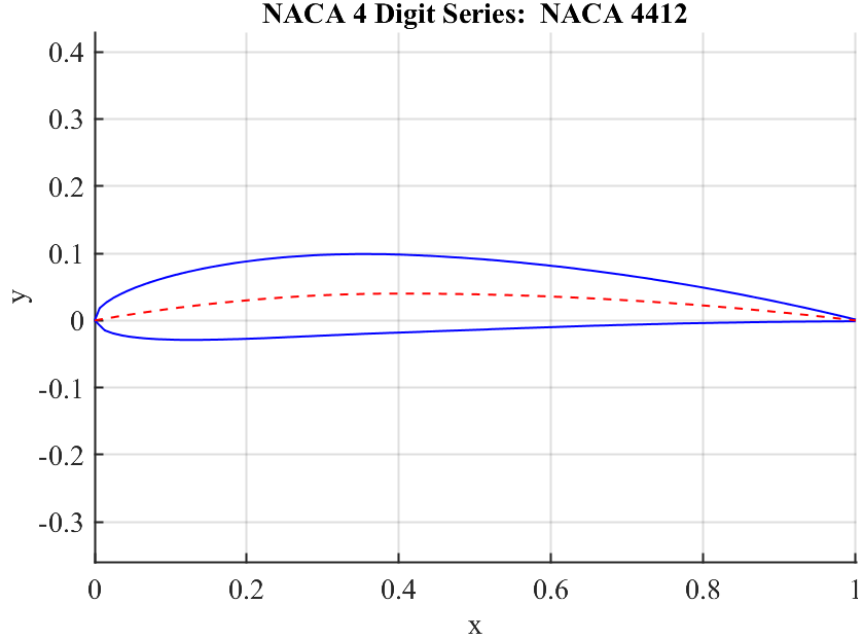
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1 Introduction

1.1 Geometrical Domain

The purpose of this exercise is to simulate the flow around the *NACA-4412* airfoil, a well-known and standardized airfoil profile. From a geometric point of view, the only data given is the length of the airfoil chord: $c = 0.1 \text{ m}$. Researches lead us to obtain data for a profile $c = 1 \text{ m}$, which is here shown:



As noticeable, the NACA 4412 is a lift wing, typically used in the aerospace industry.

In order to obtain the right coordinates for the profile required, it was necessary to scale the coordinates of the profile in the figure above. The converted values have been used to obtain the right *profile* patch in the *blockMeshDict* file. It is important to note that the airfoil features a sharp trailing edge, which was deliberately kept; this choice influenced the type of mesh used.

1.2 Physical flow

The 2D incompressible flow under study is air in standard conditions ($\nu = 15 \times 10^{-6} \text{ m}^2/\text{s}$, $\rho = 1.2 \text{ kg/m}^3$). It is given the inlet velocity: $V_{\text{in}} = 20 \text{ m/s}$, the angle of attack: $\alpha = 0^\circ$ and the ambient pressure: $P_{\text{in}} = 1 \text{ bar} \approx 101325 \text{ Pa}$. By knowing the inlet turbulence intensity $I_{\text{in}} = 1\%$ and the mixing length $l_T = 0,01\text{m}$, all the necessary parameters are given to properly set the boundary conditions of the simulations. Turbulence's effects have been modeled by employing RANS simulations that allow to simplify their study and to reduce the computational cost. For all the simulations the wing has been considered as fixed in space and with a constant AOA, so no transient simulations has been required.

2 k- ε model

2.1 Simulation Setting

A proper turbulent model must be defined. The request is to use a 2-equations model combined with wall functions; for these reasons, it has been decided to employ the **k- ε** that works well far from the walls. In particular, this case involves two additional equations and unknowns compared to a laminar simulation, necessitating the definition of suitable boundary conditions, numerical schemes, and solvers. By knowing the turbulence intensity $I_{in} = 1\%$ and the mixing length $l_T = 0,01m$ (1/10 of the chord), it has been possible to compute the inlet values for k and ε (these values are assigned to the internal field as initial guess):

$$\kappa_{in} = \frac{3}{2}U^2I^2 = 0,06m^2/s^2; \quad \varepsilon_{in} = 0.09^{3/4}\frac{\kappa^{3/2}}{l_T} = 0,2415m^2/s^3; \quad \nu_{T,in} = 0.09\frac{\kappa^2}{\varepsilon} = 0,00134m/s^2$$

BCs for k, ε, ν_T, p and U defined in the **0/** folder are summarized below:

0: BCs	k	ε	ν_T	p	U
Profile	type: kqRWallFunction value: uniform \$inletk	type: epsilonWallFunction value: uniform \$inletEpsilon	type: nutkWallFunction value: uniform 0.001342	type: zero-Gradient	type: noSlip
Inlet	type: turbulentIntensityKineticEnergyInlet intensity: 0.01 value: uniform \$inletk	type: turbulentMixingLengthDissipationRateInlet mixingLength: 0.01 value: uniform \$inletEpsilon	type: calculated value: uniform 0.001342	type: zero-Gradient	type: fixedValue value: uniform (20 0 0)
Outlet	zeroGradient	zeroGradient	zeroGradient	type: fixedValue value: uniform 0	zeroGradient
Sx / Dx	empty	empty	empty	empty	empty
Top / Bottom	symmetry	symmetry	symmetry	symmetry	symmetry

Table 2.1: 0/k, 0/epsilon, 0/nut, 0/p, 0/U

The solution algorithm for system resolution is the **SIMPLE**, and the residual control also regards the quantities k and ε . Convergence is reached when all residuals fall below 10^{-6} . Having two additional equations means that additional solvers must be defined inside the **fvSolution** file, and the discretization of the new terms in **fvSchemes**.

system/fvSchemes	Numerical scheme
ddtSchemes	steadyState
gradSchemes:grad(p)	Gauss linear
div(phi,U)	bounded Gauss linearUpwind grad(U)
div(phi,epsilon)	bounded Gauss upwind
div(phi,k)	bounded Gauss upwind
laplacianSchemes	Gauss linear corrected
interpolationSchemes	linear
snGradSchemes	corrected

Table 2.2: fvSchemes definition

system/fvSolution	solver	smoother	tolerance
p	GAMG	GaussSeidel	10^{-8}
U	smoothSolver	GaussSeidel	10^{-8}
"k—epsilon"	smoothSolver	GaussSeidel	10^{-8}

Table 2.3: fvSolution definition

The information on the turbulence model adopted is present in **constant/MomentumTransport: simulationType RAS, model kEpsilon**.

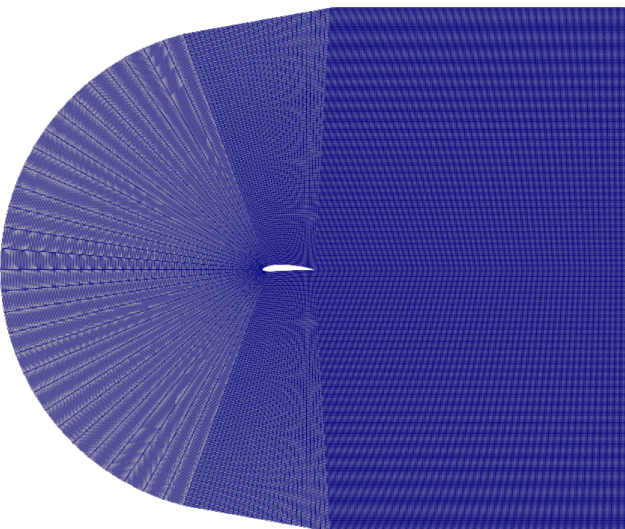
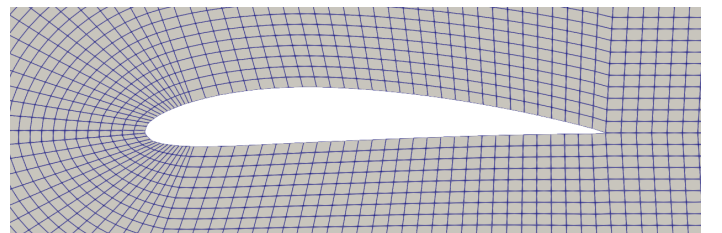
2.2 First Mesh

2.2.1 Mesh Generation

The choice of using a $C-grid$ mesh is based on the sharp trailing edge of the airfoil. It has been generated a domain that consists of five blocks surrounding the airfoil, whose dimensions have been properly given to respect the full evolution of the flow after the interaction with the airfoil and to properly set the initial boundary conditions. In fact, from a theoretical point of view, we should take into account a distance of at least $(5 \div 10) * chord$ before and after the airfoil. In this case 5 chords have been taken before the leading edge, 6 after the trailing edge. In *blockMeshDict*, a total of twenty vertices have been defined, six of which lie on the airfoil surface (3 per each side). After the definition of the five structured blocks, two circular arcs have been used to connect the inlet vertices of the domain, while six spline curves interpolate the coordinates of the points that define the airfoil profile. The seven resulting boundary patches are:

- **profile**: defined as a **wall**, representing the surface of the airfoil;
- **inlet**: patch that defines the inflow boundary;
- **outlet**: patch used for the outflow of the domain;
- **top** and **bottom**: both defined as **symmetry** planes, to model a symmetric external flow domain;
- **sx** and **dx**: both marked as **empty**, corresponding to the side faces in the spanwise direction (z), consistent with a 2D simulation.

The generated mesh has 85500 cells. It presents a *NonOrthogonality* with a maximum value of 46.36 and an average of 3.52; the max *skewness* is equal to 2.67 and the max *aspect ratio* is 8.95. **NonOrthogonalCorrector** has been added.

Figure 2.1: Coarse mesh, $k-\varepsilon$ modelFigure 2.2: Coarse mesh zoomed, $k-\varepsilon$ model

2.2.2 Results

The process has reached convergence in 299 iterations. Plots of velocity and pressure are here reported:

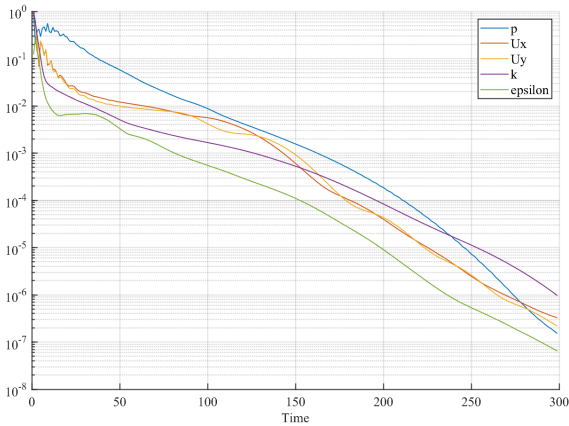


Figure 2.3: Residual trends

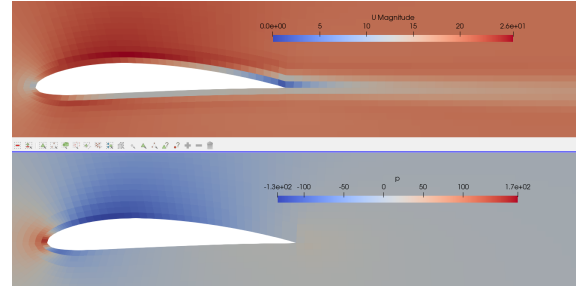


Figure 2.4: U and P

The value of y^+ must be checked. The minimum value is 41.61 and instead the maximum one is 77.71, with an average of 64.81 (all beyond the prefixed limit value, which is $y^+ = 30$). Here it is reported a graph representing y^+ located on the *profile* patch:

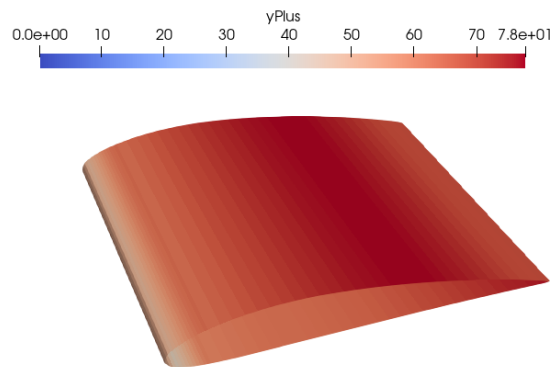


Figure 2.5: y^+ distribution along the airfoil

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,1456
C_D	0,0262
C_L	0,2934
$C_{L(f)}$	0,2924
$C_{L(r)}$	0,0010

Table 2.4: Coefficients

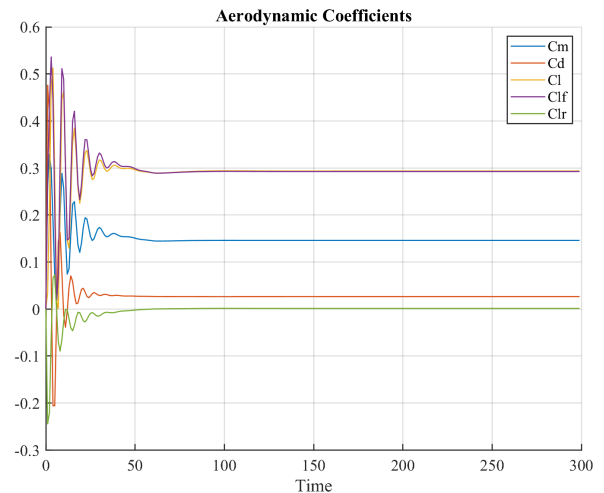


Figure 2.6: Convergence

The physics of the problem can be understood by looking at the pressure and shear stress distribution on the airfoil profile. It's evident from the first graph the difference of pressure between the two surfaces that

generates the lift. The shear distribution shows high values on the upper surface, this is probably due to the high velocity gradient linked to the reduction of the pressure.

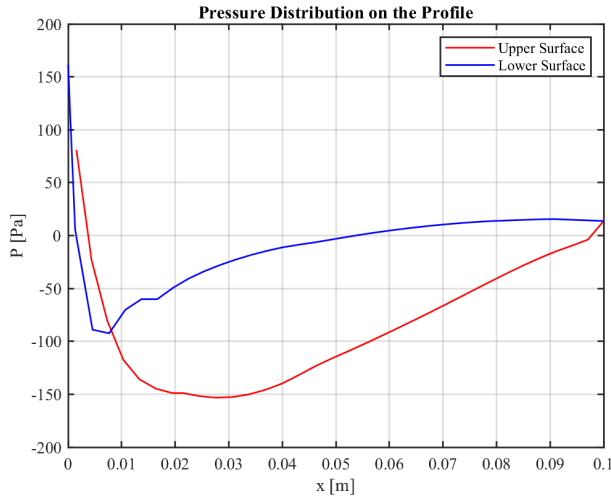


Figure 2.7: Pressure Distribution

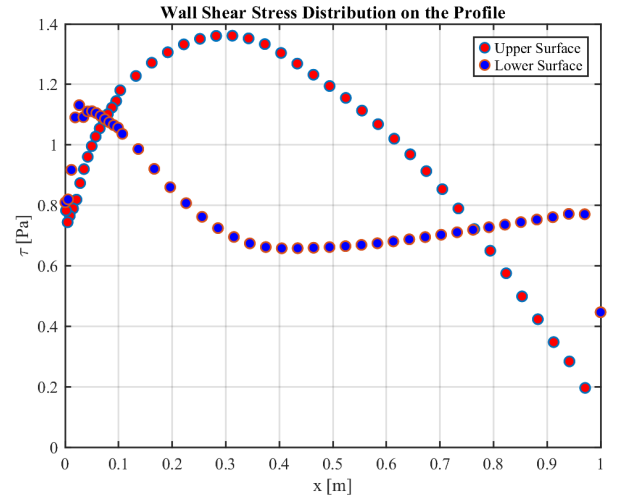


Figure 2.8: Shear Distribution

The velocity distribution at one chord downstream of the trailing edge clearly shows the wake effect that is correlated to a local decrease of the velocity:

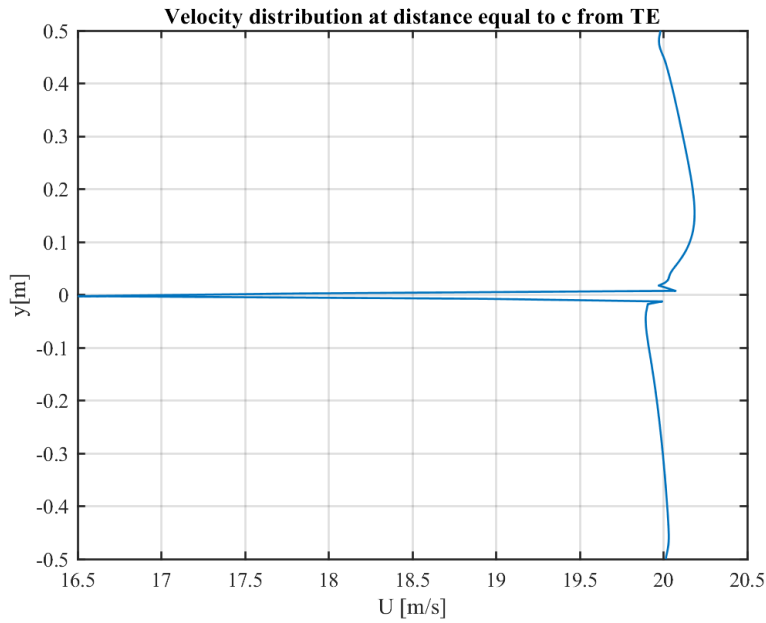
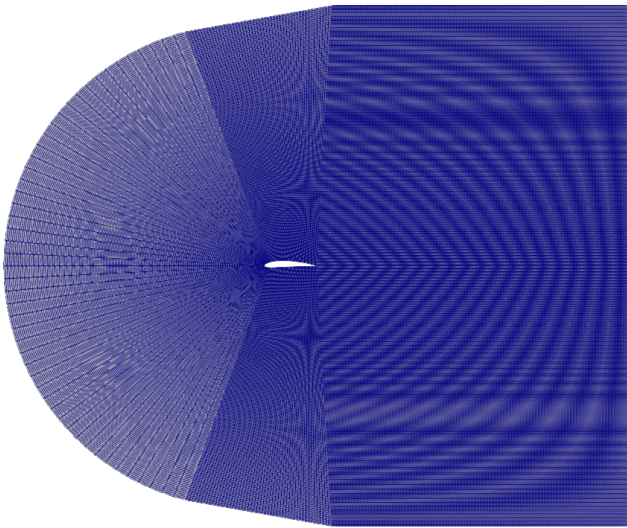
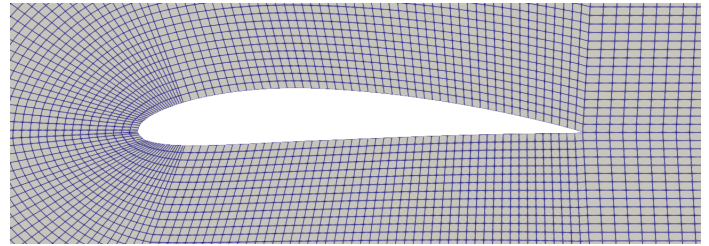


Figure 2.9: Velocity distribution in the wake

2.3 Second Mesh

2.3.1 Mesh Generation

The intermediate mesh has been generated on the same basis of the coarse one. It has been refined close to the walls through the enhancing of the gradient along the y – *direction* from 0.9 to 0.5. The number of cells has also been slightly increased: the mesh now consists of 104,500 cells. The generated mesh presents a *NonOrthogonality* with a maximum value of 48.05 and an average of 4.09; the max *skewness* is equal to 3.31 and the max *aspect ratio* is 5.71. **NonOrthogonalCorrector** has been added.

Figure 2.10: Intermediate mesh, k - ε modelFigure 2.11: Intermediate mesh zoomed, k - ε model

2.3.2 Results

The process has reached convergence in 497 iterations. Plots of velocity and pressure are here reported:

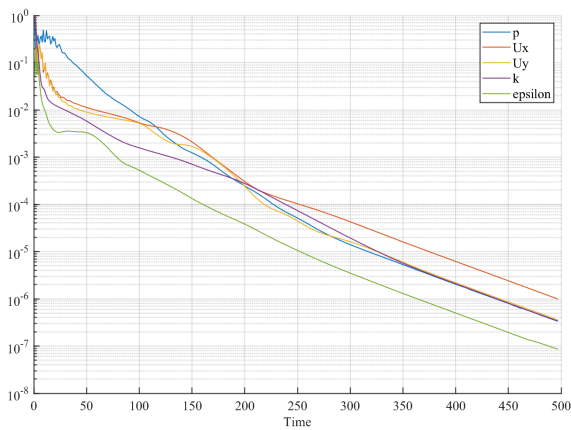


Figure 2.12: Residual trends

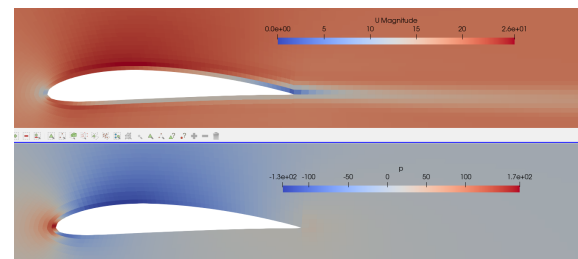
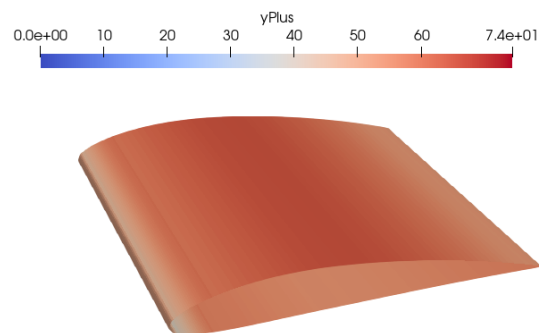


Figure 2.13: U and P

In this case the value of y^+ has as minimum 38.74 and as maximum 65.73, with an average of 56,09 (all beyond the prefixed limit value). Here it is reported a graph representing y^+ located on the *profile* patch:

Figure 2.14: y^+ distribution along the airfoil

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,1588
C_D	0,0252
C_L	0,3259
$C_{L(f)}$	0,3218
$C_{L(r)}$	0,0041

Table 2.5: Coefficients

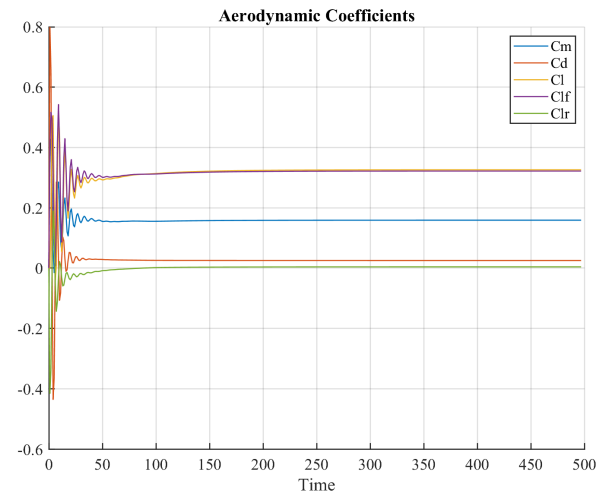


Figure 2.15: Convergence

The physics of the problem can be understood by looking at the pressure and shear stress distribution on the airfoil profile. The results show the same trend as before and the same reflections are valid.

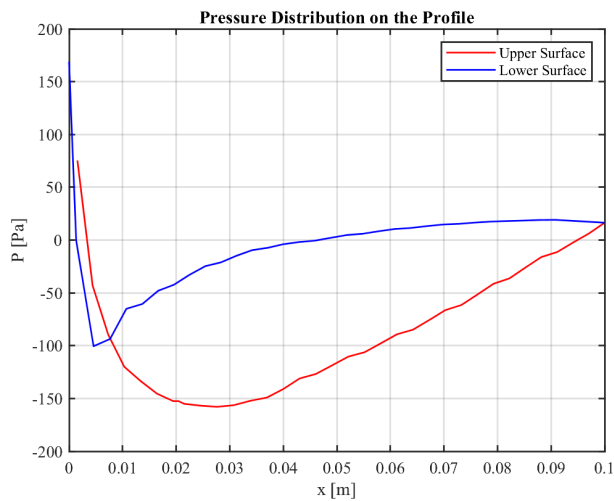


Figure 2.16: Pressure distribution

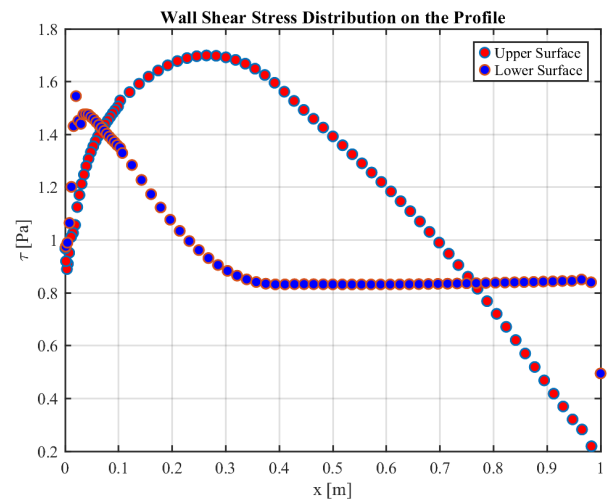


Figure 2.17: Shear distribution

The velocity distribution at one chord downstream of the trailing edge is:

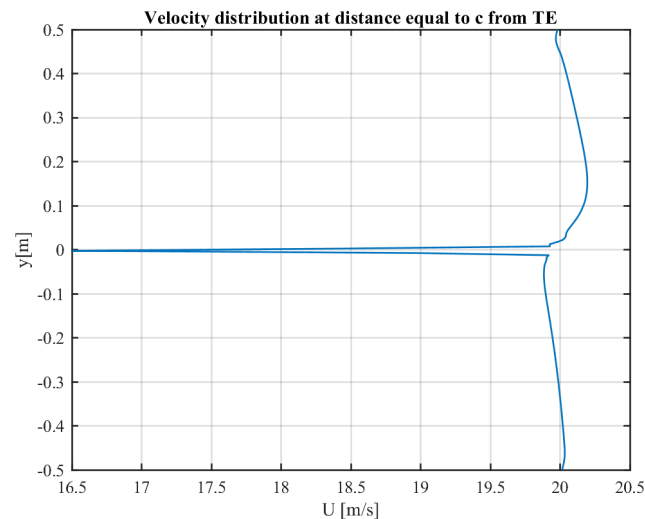


Figure 2.18: Velocity distribution in the wake

2.4 Third Mesh

2.4.1 Mesh Generation

The most refined mesh has been generated by following the same topology of the previous one, even in this case the y^+ has been decreased by enlarging the gradient along y – *direction* up to 0.4. It still has 104500 cells. The generated mesh presents a *NonOrthogonality* with a maximum value of 48.49 and an average of 4.16; the max *skewness* is equal to 2.9 and the max *aspect ratio* is 5.26.

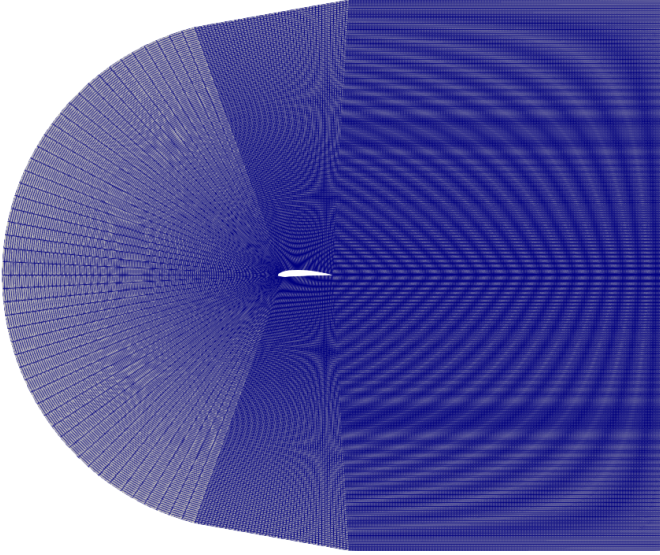


Figure 2.19: Finer mesh, k - ε model

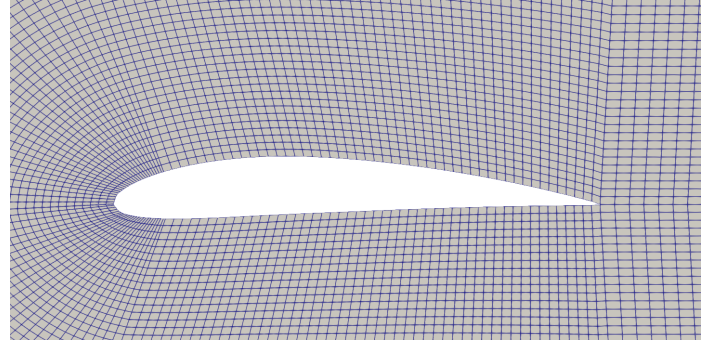


Figure 2.20: Finer mesh zoomed, k - ε model

2.4.2 Results

The process has reached convergence in 524 iterations. Plots of velocity and pressure are here reported:

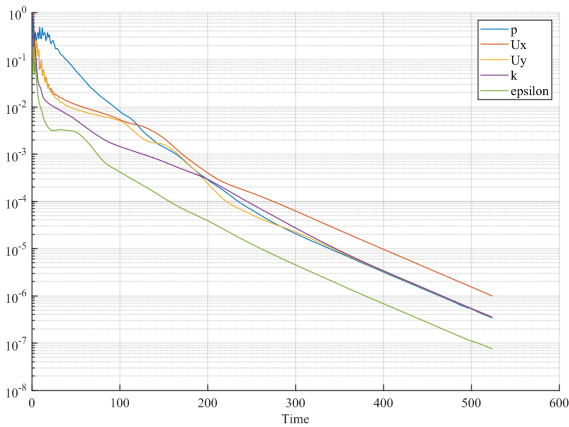


Figure 2.21: Residual trends

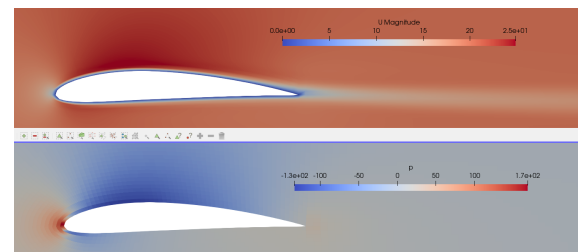
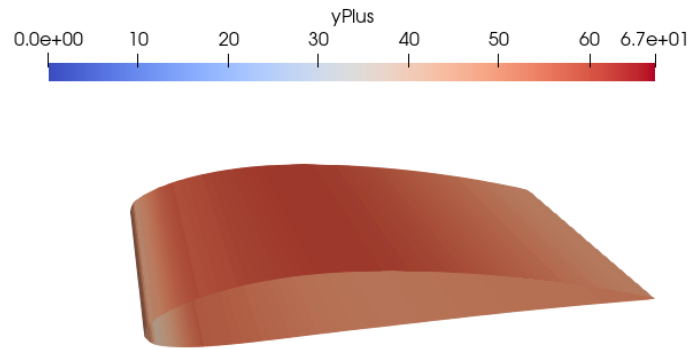


Figure 2.22: U and P

The y^+ value on the most refined mesh has a minimum value of 37.79 and a maximum value of 61.09, with an average of 52.24 (all beyond the prefixed limit value). Here it is reported a graph representing y^+ located on the *profile* patch:

Figure 2.23: y^+ distribution along the airfoil

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,1650
C_D	0,0254
C_L	0,3408
$C_{L(f)}$	0,3355
$C_{L(r)}$	0,0053

Table 2.6: Coefficients

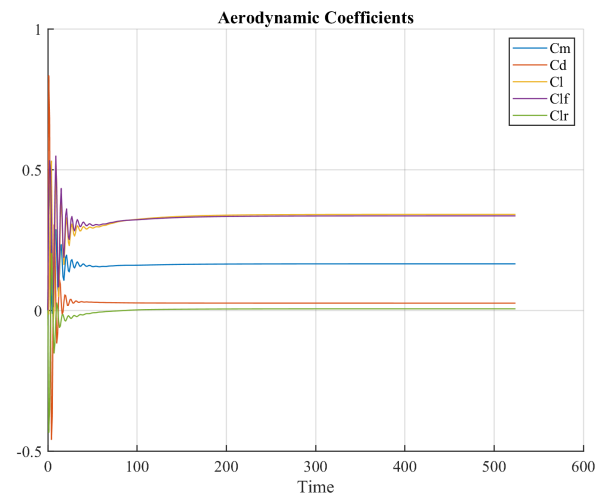


Figure 2.24: Convergence

The physics of the problem can be understood by looking at the pressure and shear stress distribution on the airfoil profile. The results show the same trend as before and the same reflections are valid.

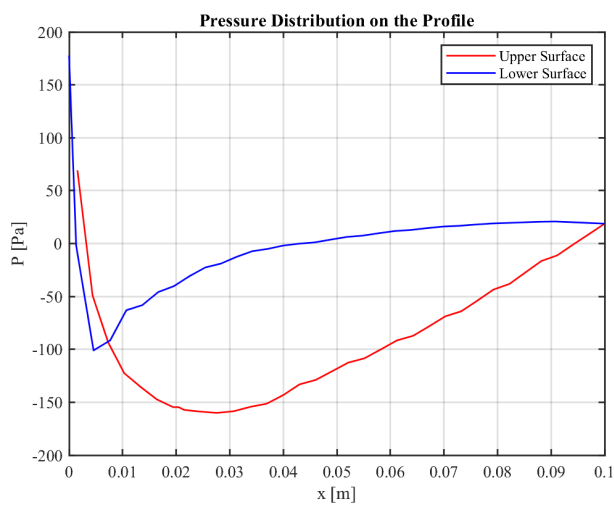


Figure 2.25: Pressure Distribution

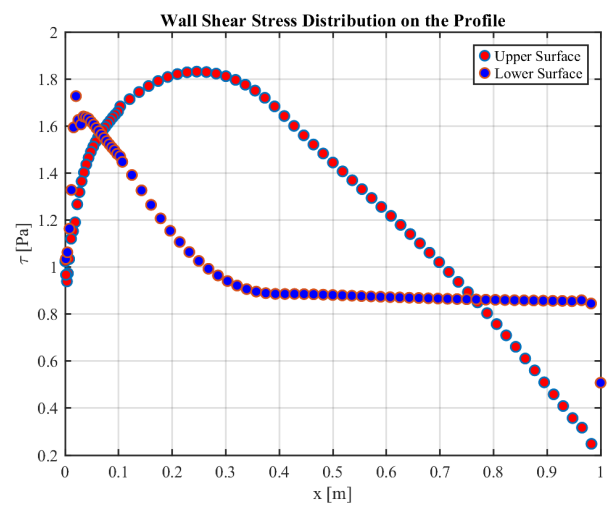


Figure 2.26: Shear Distribution

The velocity distribution at one chord downstream of the trailing edge is:

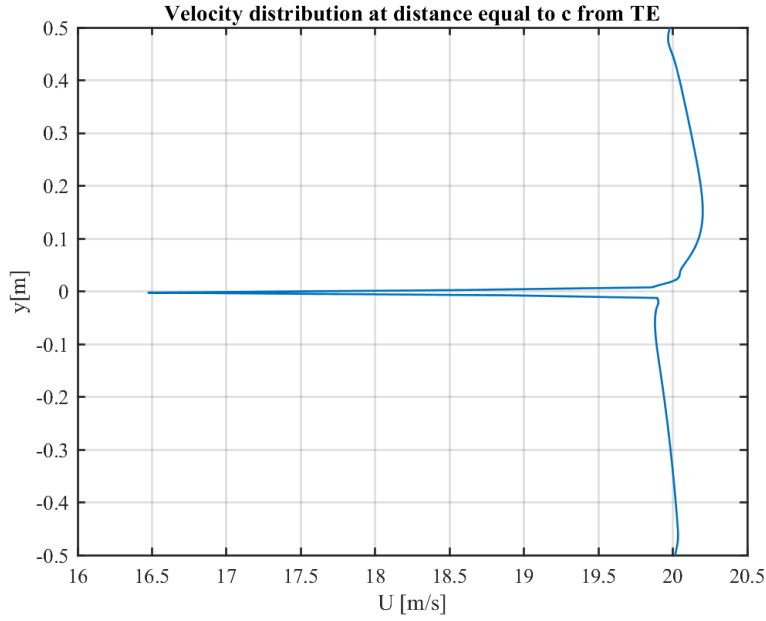


Figure 2.27: Velocity distribution in the wake

2.5 Grid Indipendence

The evaluation of the grid independence of the three simulations has been made by the comparison of the obtained results and the superimposition of the different plots in order to catch the similarities and the differences.

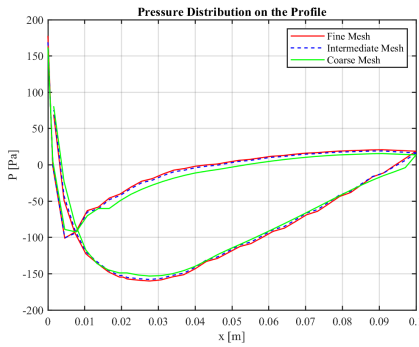


Figure 2.28: Pressure distribution

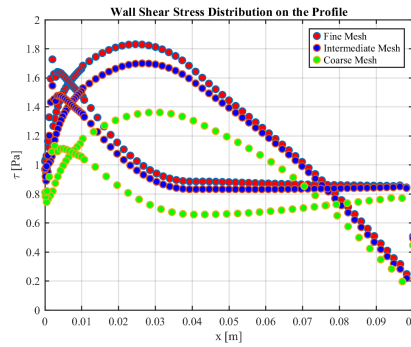


Figure 2.29: Wall shear stress distribution

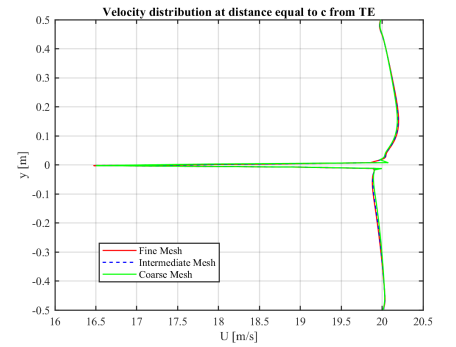


Figure 2.30: Velocity distribution, c from TE

The reported figures show that the Fine and the Intermediate meshes are almost grid independent; in fact, the plots almost completely overlap. The solution appears to be sensitive to mesh resolution, particularly in the wall shear stress distribution, where more noticeable differences can be observed among the three meshes, this is probably due to direct dependence on the velocity gradient that is strictly dependent on the level of refinement close to the walls. For sure better results would be achieved by lowering the y^+ of the mesh but in this case same meshes have been employed also for the simulations with *SA* model and this represented a limit as it will be explained in the next chapter.

3 Spalart-Allmaras model: Wall Function

3.1 Simulation Setting

The other request of the exercise was to perform the simulations using the SA model. In this case there is only the balance of the *eddy viscosity* μ_T as an additional equation to close the turbulence model. Consequentially the boundary conditions (0/), except for p and U, the solvers (system/fvSolutions) and the discretization schemes (system/fvSchemes) are different from the previous case. The flow is in the same physical condition.

0: BCs	$\tilde{\nu}$	ν_T
Profile	<code>type: fixedValue</code> <code>value: uniform 0.001342</code>	<code>type: nutUSpalding-WallFunction</code> <code>value: uniform 0.001342</code>
Inlet	<code>type: fixedValue</code> <code>value: uniform 0.001342</code>	<code>type: calculated</code> <code>value: uniform 0.001342</code>
Outlet	<code>zeroGradient</code>	<code>zeroGradient</code>
Sx / Dx	<code>empty</code>	<code>empty</code>
Top / Bottom	<code>symmetry</code>	<code>symmetry</code>

Table 3.1: 0/nuTilda, 0/nut

The solution algorithm for system resolution is the **SIMPLE**, and in this case the residual control also regards the quantity *nuTilda*. Convergence is reached when all residuals fall below 10^{-5} .

system/fvSchemes	Numerical scheme
ddtSchemes	<code>steadyState</code>
gradSchemes:grad(p)	<code>Gauss linear</code>
div(phi,U)	<code>bounded Gauss linearUpwind grad(U)</code>
div(phi,nuTilda)	<code>bounded Gauss linearUpwind grad(nuTilda)</code>
laplacianSchemes	<code>Gauss linear corrected</code>
interpolationSchemes	<code>linear</code>
snGradSchemes	<code>corrected</code>

Table 3.2: fvSchemes definition

system/fvSolution	solver	smoother	tollerance
p	<code>GAMG</code>	<code>GaussSeidel</code>	10^{-8}
U	<code>smoothSolver</code>	<code>GaussSeidel</code>	10^{-8}
nuTilda	<code>smoothSolver</code>	<code>GaussSeidel</code>	10^{-8}

Table 3.3: fvSolution definition

The information on the turbulence model adopted is present in **constant/MomentumTransport: simulationType** RAS, model SpalartAllmaras.

The aim of the exercise is to compare the results of the two models $k - \varepsilon$ and *SA*, for this reason the same meshes previously described have been adopted. In the next paragraphs reference will be made to them without reporting the images again. It is important to say that the minimum y^+ values obtained with *SA* are lower with respect to $k - \varepsilon$ and this represented the constraint on the mesh generation that brought to have y^+ values above the threshold in the 2 equation model simulations.

3.2 First Mesh

3.2.1 Results

The process has reached convergence in 862 iterations. Plots of velocity and pressure are here reported:

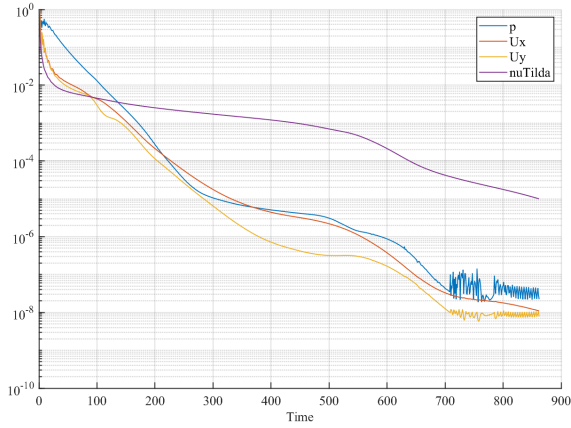


Figure 3.1: Residual trends

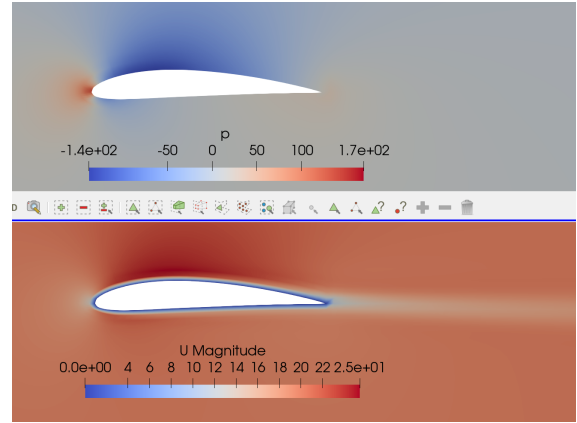


Figure 3.2: U and P

The y^+ has a minimum value of 41,95 and a maximum value of 113,28, with an average of 83,11. Here it is reported a graph representing y^+ located on the *profile* patch:

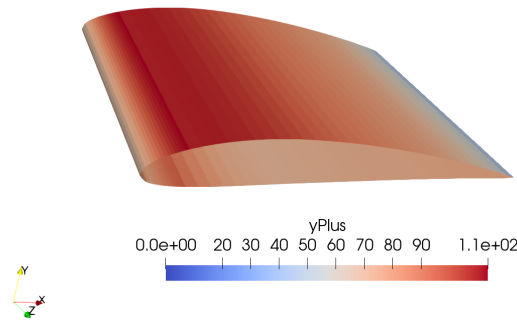


Figure 3.3: y^+ distribution along the airfoil

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,1814
C_D	0,0261
C_L	0,3795
$C_{L(f)}$	0,3712
$C_{L(r)}$	0,0058

Table 3.4: Coefficients

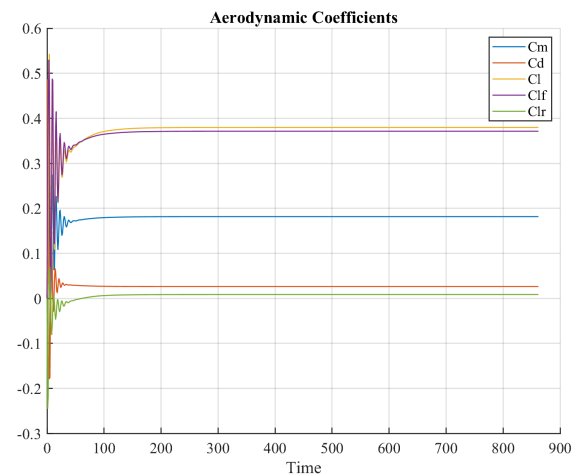


Figure 3.4: Convergence

The pressure and shear stress distribution on the airfoil profile obtained in this case are:

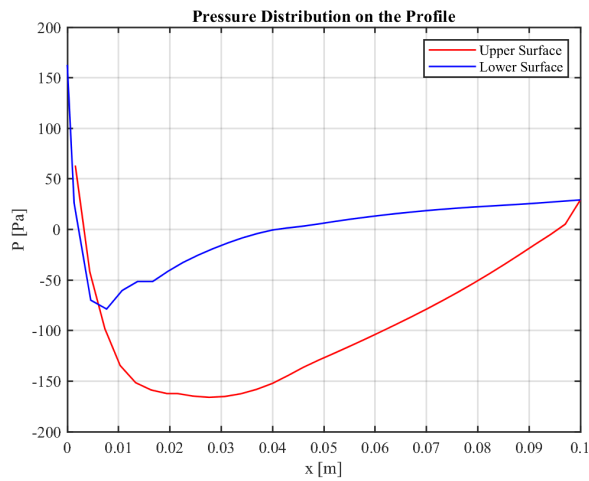


Figure 3.5: Pressure Distribution

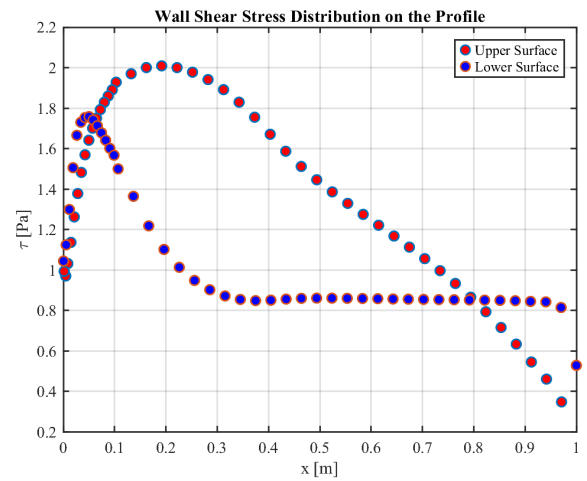


Figure 3.6: Shear Distribution

The velocity distribution at one chord downstream of the trailing edge is:

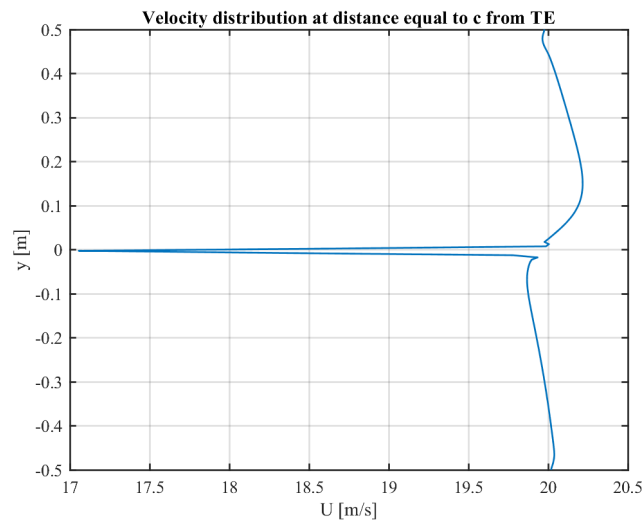


Figure 3.7: Velocity distribution in the wake

3.3 Second Mesh

The process has reached convergence in 968 iterations. Plots of velocity and pressure are here reported:

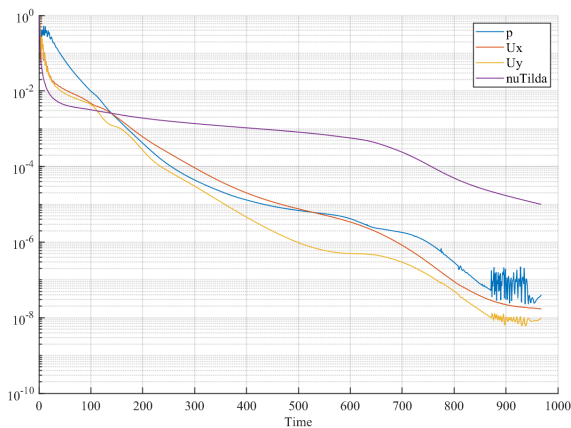


Figure 3.8: Residual trends

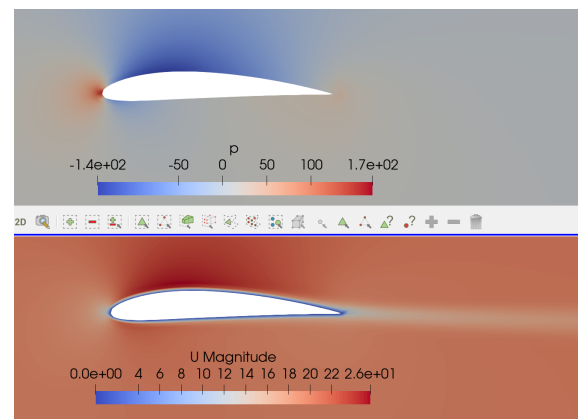


Figure 3.9: U and P

The y^+ has a minimum value of 33,47 and a maximum value of 85,82, with an average of 63,85. Here it is reported a graph representing y^+ located on the *profile* patch:

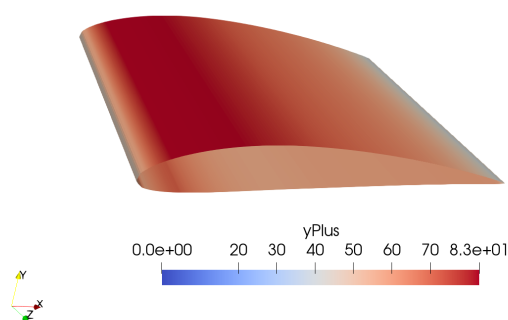


Figure 3.10: y^+ distribution along the airfoil

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,1902
C_D	0,0247
C_L	0,4017
$C_{L(f)}$	0,3911
$C_{L(r)}$	0,0106

Table 3.5: Coefficients

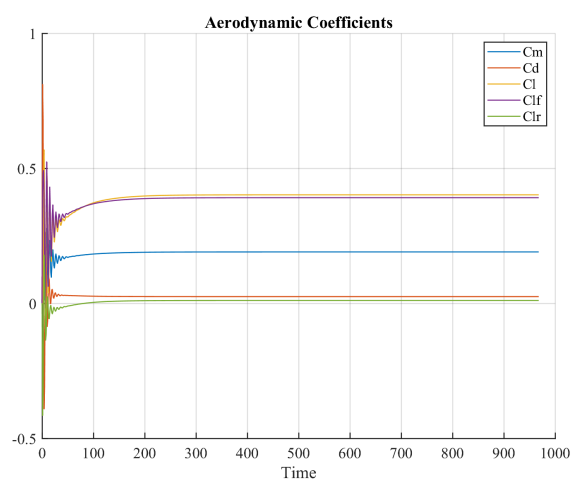


Figure 3.11: Convergence

The pressure and shear stress distribution on the airfoil profile obtained in this case are:

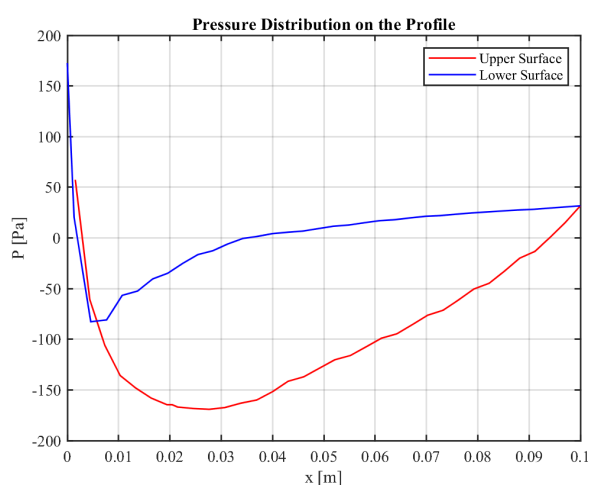


Figure 3.12: Pressure Distribution

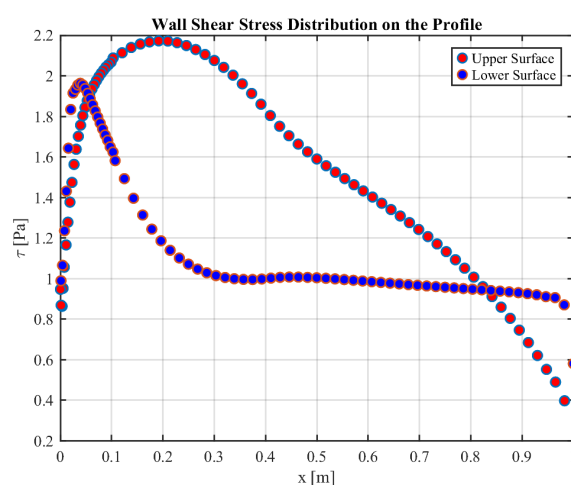


Figure 3.13: Shear Distribution

The velocity distribution at one chord downstream of the trailing edge is:

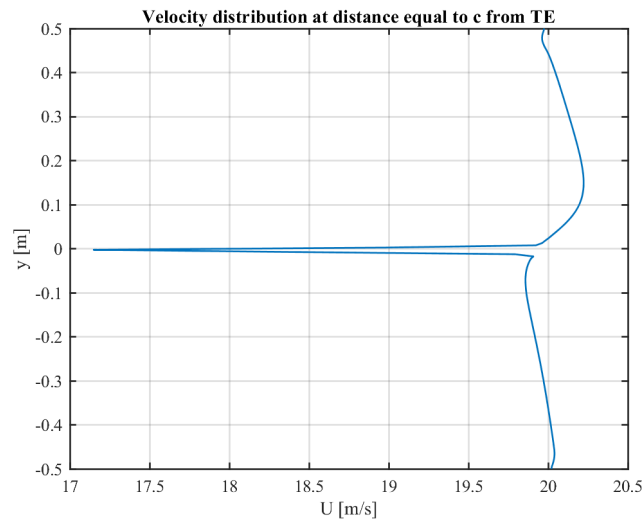


Figure 3.14: Velocity distribution in the wake

3.4 Third Mesh

The process has reached convergence in 964 iterations. Plots of velocity and pressure are here reported:

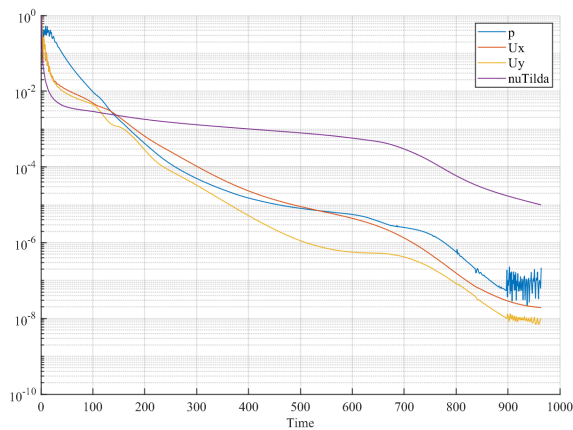


Figure 3.15: Residual trends

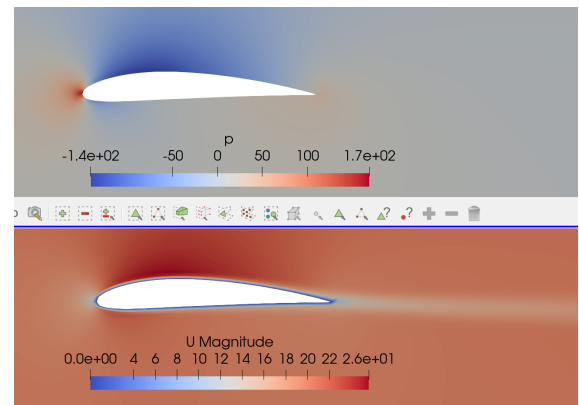
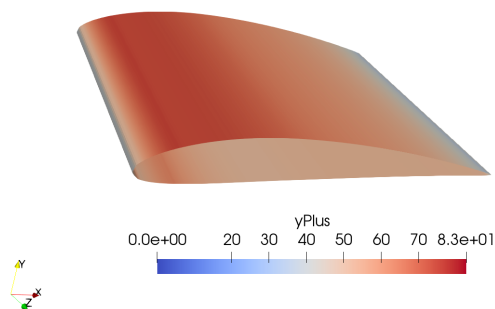


Figure 3.16: U and P

The y^+ has a minimum value of 30,41 (near the threshold) and a maximum value of 76,49, with an average of 57,16. Here it is reported a graph representing y^+ located on the *profile* patch:

Figure 3.17: y^+ distribution along the airfoil

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,1933
C_D	0,0248
C_L	0,4090
$C_{L(f)}$	0,3978
$C_{L(r)}$	0,0111

Table 3.6: Coefficients

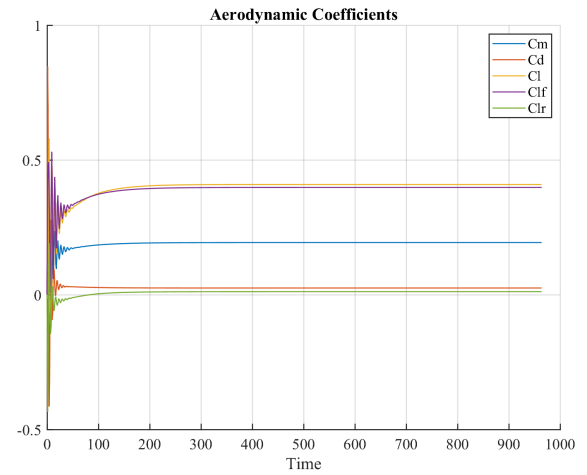


Figure 3.18: Convergence

The pressure and shear stress distribution on the airfoil profile obtained in this case are:.

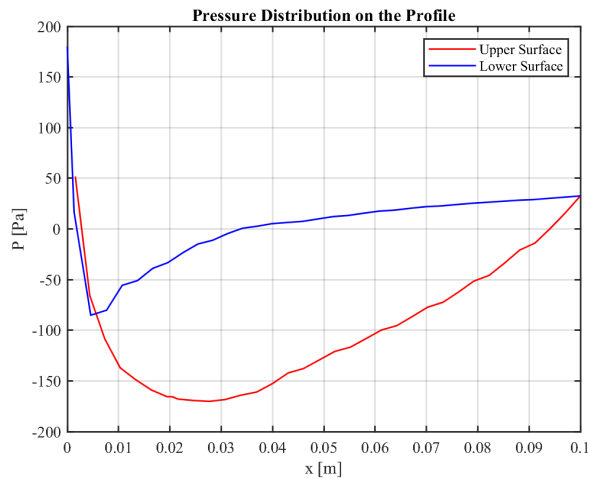


Figure 3.19: Pressure Distribution

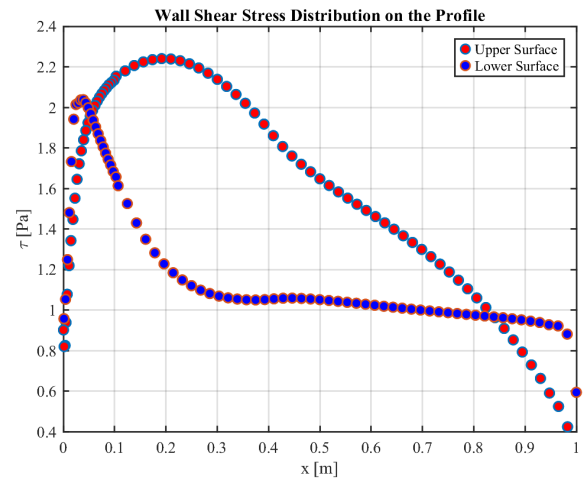


Figure 3.20: Shear Distribution

The velocity distribution at one chord downstream of the trailing edge is:

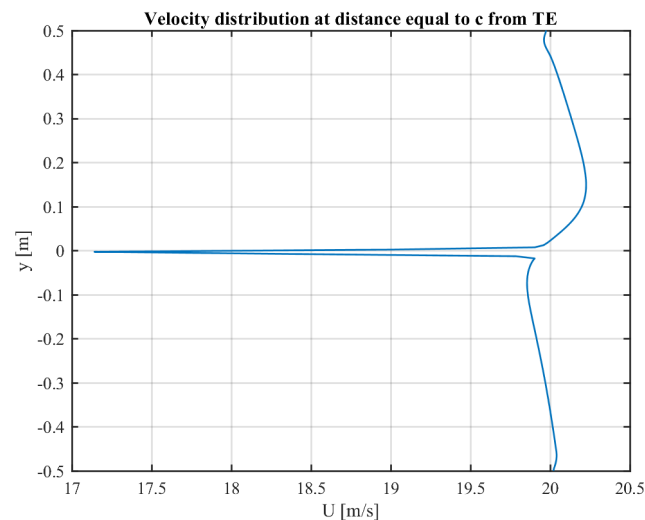


Figure 3.21: Velocity distribution in the wake

3.5 Grid Independence

Even in this case the evaluation of the grid independence of the three simulations has been made by the comparison of the obtained plots:

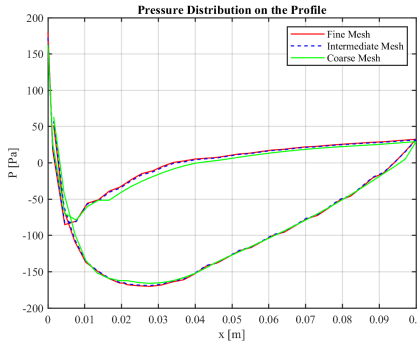


Figure 3.22: Pressure distribution

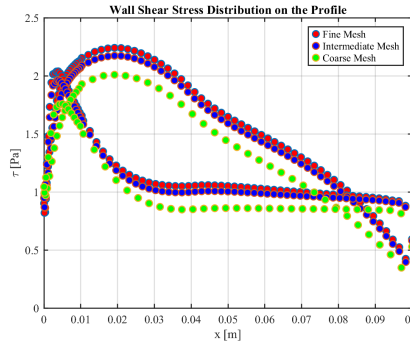


Figure 3.23: Wall shear stress distribution

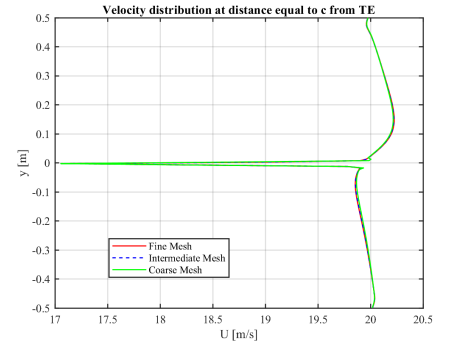


Figure 3.24: Velocity distribution, c from TE

The three simulations give very similar results, in particular the fine and the intermediate ones seem to be almost grid independent. With respect to $k - \epsilon$ the relative error is lower and this is probably due to the reduced and more similar values of y^+ of the three meshes.

3.6 SA and $k - \epsilon$ comparison

Here is reported the comparison of the results obtained using the two different turbulence models on the same mesh.

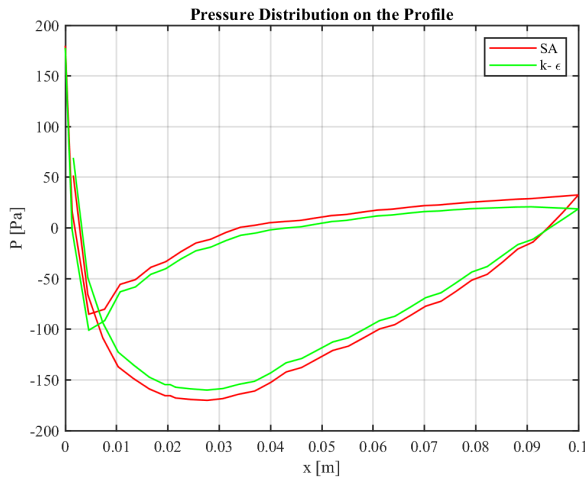


Figure 3.25: Pressure Distribution

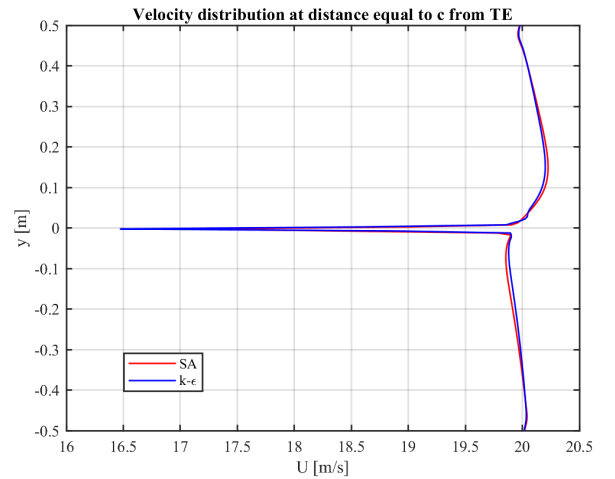


Figure 3.26: Velocity Distribution

It is clear from the images that both models give coherent results which slightly differ. The largest discrepancy occurs in the pressure distribution, as this region is most affected by the presence of the wall and, consequently, by the modeling approach and the adopted y^+ values.

4 Spalart-Allmaras model: No WF

4.1 Simulation Setting

The same boundary conditions used in the previous SA case with wall functions are applied here, as OpenFOAM can automatically determine whether wall functions are necessary or not. The only difference is that to the value of ν_T on the profile patch has been assigned $1e - 12$ as initial guess for *nutUSpaldingWallFunction*. The same numerical schemes and solution algorithms described earlier are also employed. In this case as well, convergence is considered achieved when all residuals fall below 10^{-5} in order to avoid an excessively heavy simulation.

4.2 Mesh Generation

The mesh that has been used it is an evolution of the previous ones. As it can be seen in the figures below, now the blocks implied in the mesh are 10. It has been decided to have a thin layer surrounding the wing of 5 blocks, such as to have a strong grading in the y-direction (0,008) and yet have a light mesh, with a sustainable number of cells. To further reduce the mesh weight compared to previous simulations, one chord length was removed from the rear portion of the mesh (keeping still five chords originating from the trailing edge to achieve reliable results). These measures made it possible to generate a mesh with appropriate y^+ values to avoid the use of wall functions, while also reducing the total number of cells to 45500.

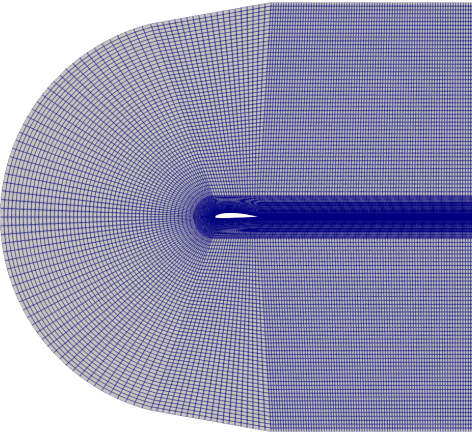


Figure 4.1: Mesh, SA model, no WF

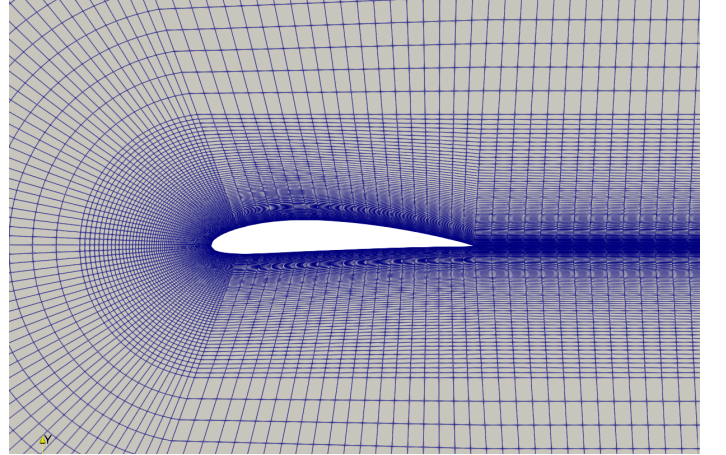


Figure 4.2: Mesh zoomed, SA model, no WF

It presents a *NonOrthogonality* with a maximum value of 54.25 and an average of 10.39; the max *skewness* is equal to 0.92 and the max *aspect ratio* is 406.053. The mesh topology presented was adopted as it offered the best compromise among the quality indexes discussed above.

4.3 Results

The process has reached convergence in 11773 iterations. Plots of velocity and pressure are here reported:

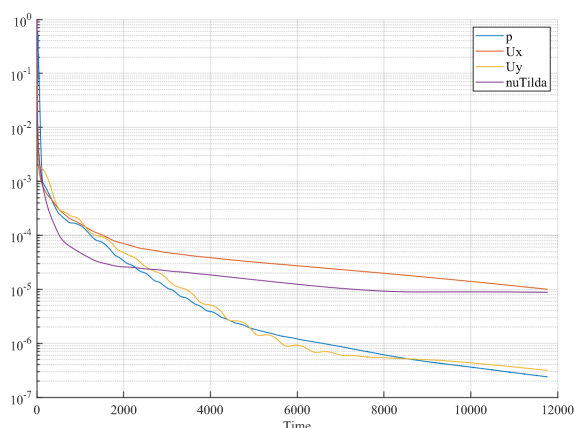


Figure 4.3: Residual trends

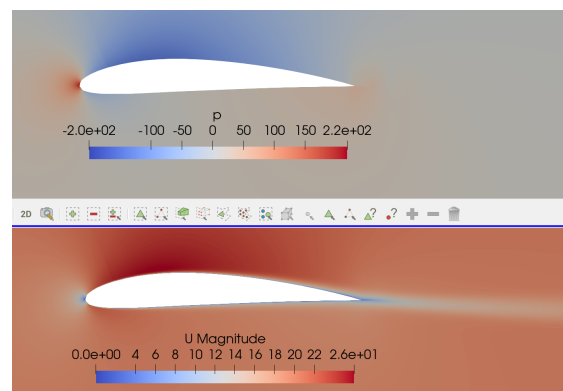


Figure 4.4: U and P

The y^+ has a minimum value of 0,102 and a maximum value of 1,33, with an average of 0,688. Here it is reported a graph representing y^+ located on the *profile* patch:

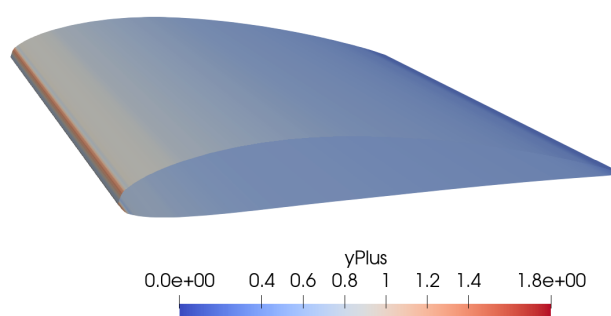


Figure 4.5: y^+ distribution along the airfoil

There is only a small region near the leading edge, where the boundary layer is still developing, that has a y^+ value that slightly exceeds the common threshold 1, this doesn't represent a problem for the validity of the results.

The aerodynamic coefficients relating to the performed simulation are:

Coefficient	Value
C_m	0,2420
C_D	0,0250
C_L	0,5089
$C_{L(f)}$	0,4965
$C_{L(r)}$	0,0124

Table 4.1: Coefficients

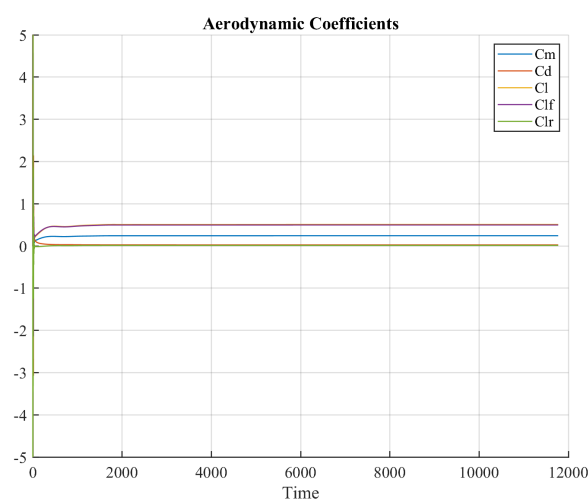


Figure 4.6: Convergence

The pressure and shear stress distribution on the airfoil profile obtained in this case are:

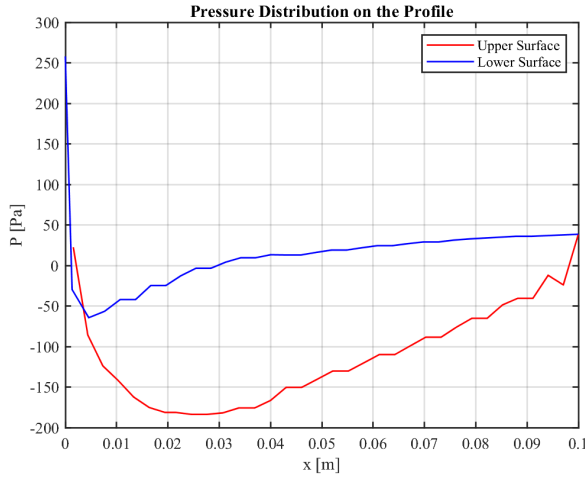


Figure 4.7: Pressure Distribution

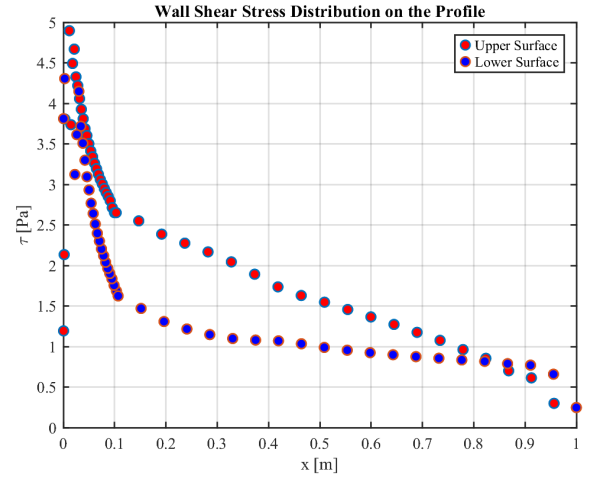


Figure 4.8: Shear Distribution

The pressure distribution seems to be less smooth than the one obtained with the employment of wall functions, this is probably linked to the higher sensitivity to the position of the pressure probes. The overall behavior of the stress distribution is the same of the WF cases but here there are higher values close to the leading edge.

The velocity distribution at one chord downstream of the trailing edge is:

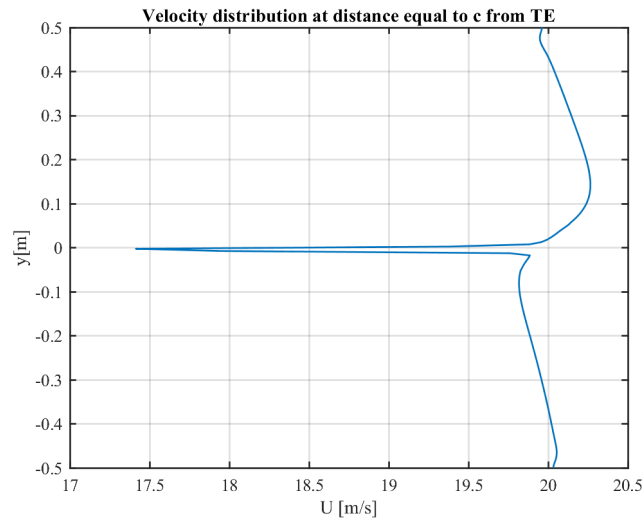


Figure 4.9: Velocity distribution in the wake

The plots seem to be coherent with the results of the other simulations so it is possible to affirm that overall description of the flow is correct, except for small errors in the values of the computed variables.

5 Conclusions

Here are presented some considerations about the choices made for the simulations and the results obtained:

- For the discretization of the divergence term $div(phi, U)$ different solutions have been tested such as: *Upwind*, *Linear*, *LimitedLinear*. The best one in terms of convergence speed and quality of the results resulted to be the *linearUpwind Grad(U)*;
- Dealing with a steady state case for the divergences numerical schemes, the *bounded* versions have been used. It increases the Diagonal Dominance of the solution matrix making the equation less difficult to solve;
- For the discretizations of the divergence terms $div(phi, epsilon)$ and $div(phi, k)$ it has been decided to use *Gauss Upwind*, because surely stable even if less accurate. The focus on accuracy was mainly on the physical variables p and U .
- Regarding the results, from a physical point of view, the typical behavior of a flow around a lift wing is present in all simulations. A positive Cl coefficient is obtained: the flow gives a clear push from bottom to top, due to the delta pressure between the pressure and suction sides.
- Analyzing the Cl results for the 3 different cases (fine mesh for k-epsilon, fine mesh for SA with wall-functions and the mesh for SA without wall-functions), they show differences non negligible. Among the first two cases, Cl changes from 0,3408 to 0,4090. The difference can be related to the use of a different model with different values of y^+ . The third case has a Cl equal 0,5089. In this case no wall-functions are implied, so it may be that this result is more reliable. Nonetheless, appropriate experimental validation procedures and further analyses may still be required to ensure robustness of the results.